Thesis_v0.9

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Abstract

Keywords

Introduction

Methods

Based on Okunogbe A, Nugent R, Spencer G, et al, this paper asseses the current and future impacts of overweight and obesity between 2020 and 2060. More specifically, a cost-of illness approach as described in D P Rice in "Estimating the cost of illness.", splitting economic costs into direct and indirect costs is applied. This methodology is used to arrive the impact of obesity in monetary terms. It is useful for understanding the detrimental effects of obesity in terms of agenda-setting and policy prioritization.
The analysis includes estimates until 2050 and 2060, following the approach of Okunogbe A, Nugent R, Spencer G, et al. Estimations are done on a country-level basis of in total 161 countries. Data is sourced from publicly available databases, detailed information on the respective data and calculations is described in the corresponding section. A comprehensive overview of the data used can be seen in Table 1.1.@ref{tab: test}

Projection_Type	Future_Parameter_Values	Source
Projection from secondary sources	Population	United Nations Population Division (UNPD) (projections to 2060)[28]
Projection by authors	Total health expenditure per capita	Chang et al, 2019 (Global Burden of Disease Health Financing Collaborator Network)[31] (projections to 2050)
Projection from secondary sources	All-cause mortality for Obesity related diseases	Foreman et. al, 2019. (projections to 2040)[32]

Projection_Type	Future_Parameter_Values	Source
Projection by authors	Life expectancy	United Nations Population Division (projections to 2060)[28]
Projection from secondary sources	Background death rates	United Nations Population Division (projections to 2060)[28]
Projection by authors	Annual GDP for Australia, Brazil, India, Mexico, Saudi Arabia, South Africa, Spain	OECD Economic Outlook - Long-term baseline projections (projections to 2060)[33]
Projection from secondary sources	Annual GDP for Thailand	World Bank World Development Indicators[34]
Projection by authors	GDP Deflator for Australia, Brazil, India, Mexico, Saudi Arabia, South Africa, Spain	OECD Economic Outlook (Long-term baseline projections to 2060) [33]
Projection from secondary sources	GDP Deflator for Thailand	World Bank World Development Indicators[34]
Projection by authors	Labor Force Size and Employment Rates for Australia, Brazil, India, Mexico, Saudi Arabia, South Africa, Spain	OECD Economic Outlook (Long-term baseline projections to 2060) [33]
Projection from secondary sources	Labor Force Size and Employment Rates for Thailand	International Labour Organization (ILO)[35]
Projection by authors	Overweight and Obesity prevalence for ages under and above 20 years and by sex.	NCD-Risk Factor Collaboration (1975-2016)[36]
Projection from secondary sources	Obesity Attributable Fraction (OAF) of Health Expenditures.	Current OAF estimations from OECD study[5] and projected prevalence.
Projection by authors	Wages	ILO Global Wage Report[37], OECD[38]
Projection from secondary sources	Assumed to stay constant (financial parameters adjusted for inflation)	Assumed to stay constant (financial parameters adjusted for inflation)

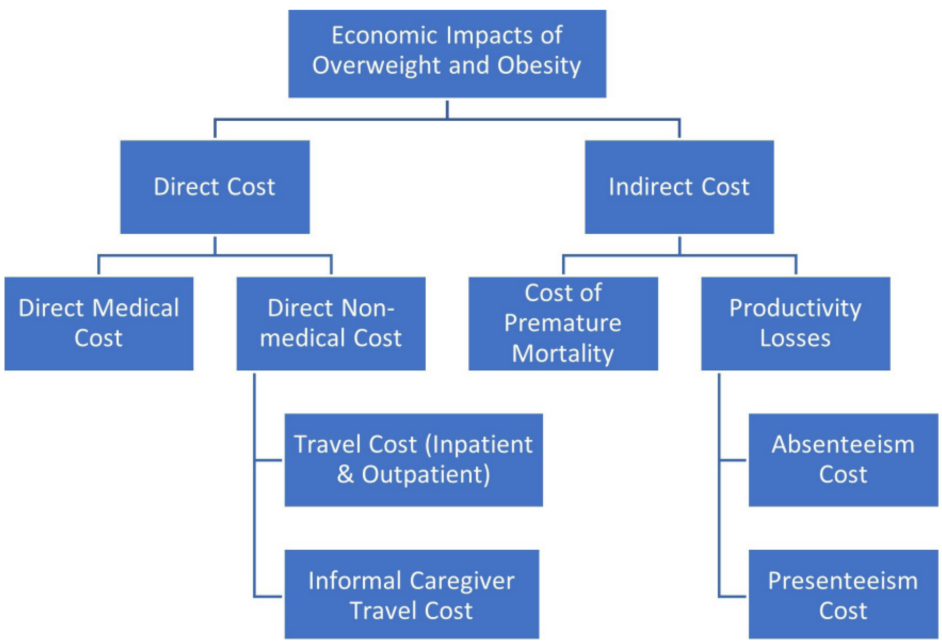
Projection_Type	Future_Parameter_Values	Source
Projection by authors	Average travel cost to and from health facility; Average number of inpatient consultations by population with obesity; Average number of hospitalization days of population with obesity; Average number of outpatient consultations by population with obesity; Absenteeism days; Presenteeism rate	

INSERT TABLE

By utilizing the cost-of-illness approach, the impacts of overweight and obesity are divided into direct and indirect cost. This thesis shall explain these costs and the performed calculations in a systematic manner, conveying a complex issue and extensive calculations in a comprehensive but also comprehensible way. The paper is therefor divided into 3 sections:

1. Overweight and obesity prevalence
2. Direct costs
3. Indirect costs

The paper starts by inspecting the current situation of obesity prevalence and projecting this into 2060 by fitting autoregressive models to forecast future behavior based on past behavior data. We can then move on to direct and indirect costs, starting by closely inspecting direct costs. These include direct medical costs and direct non-medical costs. Indirect costs will be treated as the last part and consist of costs due to premature mortality and costs due to absenteeism and presenteeism.



INSERT PICTURE OF ARCHITECTURE

Overweight and Obesity

Calculating any type of cost related to overweight and obesity requires data on overweight and obesity prevalence. It is therefore a central object of this paper to predict future obesity prevalence in order to later use these predictions to arrive with estimates of the total economic costs related to overweight and obesity.

Data

Data from NCD Risk Collaboration group is sourced as a main basis for projecting future overweight and obesity prevalence. More specifically, this data includes country-level annual prevalence values for the years 1975 to 2016. Notably, these values encapsulate all BMI categories. We thus aggregate these categories to form a comprehensive perspective referred to as “overweight and obesity.” In the context of this paper, the term “overweight and obesity” refers to BMI values above the established overweight threshold of $25, \mathrm{kg/m}^2$. To ensure a nuanced and detailed analysis, projections are made by age group and sex. Data from NCD Risk Collaboration thus includes one dataset for all males (referred to as men) and females (women) above 20 years of age, and one dataset for males (boys) and females (girls) below 20 years of age. Each group´s prevalence is projected separately.

Furthermore, besides Age and Sex, Okunogbe A et al. identified *Demographic transition* and *Nutrition transition* as proxies for drives of trends in overweight and obesity prevalence.
Demographic transition is presented by the proportion of the total population in aggregated age groups: ‘under 20 years’, ‘20-54 years’, ‘54-74 years’ and ‘above 75 years’. We shall therefore denote it *demographic change* when modelling. Data on Population is sourced from United Nations Population Divisions, including past values from 1975-2016 and projections until 2060.
Nutrition transition is approximated by *urbanization*, measured as the percentage of population living in urban areas. As population tends to shift from rural to urban environments, there is also a shift in dietary habits. This dimension should be captured by incorporating urbanization into our model.

However, after importing the data from the United Population Division it appears that contrary to Okunogbe et al., data on urbanization is only available until 2050. Unfortunately, this limits the projections of the developed time series model as we need data until 2060 to project overweight and obesity prevalence until 2060 with the corresponding covariate.
The subsequent section will explain the modelling approach, creating an understanding of how the described factors are used in modelling and why urbanization sadly cannot be incorporated.

Modelling and Projections

Following the approach of Okunogbe A et al. in “Economic impacts of overweight and obesity: current and future estimates for eight countries” we estimated a multivariate autoregressive model. We intend to do this on a global scale, for 161 countries, extending the geographical aspect in the paper “Economic impacts of overweight and obesity: current and future estimates for eight countries”.

The model employed retains the same structure as the one proposed by Okunogbe et al., taking the following form for *country i, year t and sex s*:

$$\Delta Y_{i,s,t} = \alpha + \sum_{h=1}^{n} \beta_h \Delta Y_{i,s,t-h} + X'_{i,s,t-1}\theta + \epsilon_{i,s,t}$$

where:

- $\Delta Y_{i,s,t}$ represents the differenced transformation of obesity prevalence with n degrees of lag.
- $X_{i,s,t}$ are external factors influencing obesity prevalence. That is, demographic changes and urbanization in our case. For demographic changes (Population) the age group ‘under 20 years’ was used as exogenous variable for boys and girls, and for male and female the remaining age groups(20-52 years, 54-74 years, ‘above 75 years’)
- α is the intercept.
- $\epsilon_{i,s,t}$ refers to the error term.

With the intention of producing reasonable values of obesity prevalence for future years and trying to reproduce the modelling approach by Okunogbe et al.,we decided to stick with the same autoregressive order, degree of differencing, and external factors influencing the time series data. However, for external factors we could sadly only use population change and had to dismiss the use of urbanization due to the lack of data.

The chosen lag of 2 degrees in the autoregressive component and first-order differencing of all variables ensures that our model accounts for past prevalence observations and captures the dynamics of demographic changes.

In the end, the model takes the following form:

$$\Delta Y_{i,s,t} = \alpha + \sum_{h=1}^{2} \beta_h \Delta Y_{i,s,t-h} + X'_{i,s,t-1}\theta + \epsilon_{i,s,t}$$

(Okunogbe et al. 2021)

Furthermore, time series data (prevalence of overweight and obesity) is constraint to an upper bound of 100% using a logit transformation as expressed by the formula

$$\log\left(\frac{\text{overweight\_obesity}}{1 - \text{overweight\_obesity}}\right).$$

Values are then re-converted back before forecasting the prevalence in such a way, that forecasts in the end consist of 'real' numbers.

The intention is to use this for all 161 countries, but this is unfortunately not possible due to the nature of the data. While fitting the model was not a problem in most cases, there are some 'outliers' like Australia or Papua New Guinea for which stationary was not fulfilled. In these cases we implemented a second order of differencing to ensure stationarity. This adjustment not only facilitated successful model fitting and predictions but also mitigated errors like the one encountered, specifically *Error in* `stats::arima(x = x, order = order, seasonal = seasonal, xreg = xreg)`: *non-stationary AR part from CSS*.

It is worth noting that the model parameters were set consistently with the specifications outlined in the reference paper that served as the basis for this analysis. Interestingly, the reference paper did not explicitly mention encountering any errors during their modeling process. While the model process is extended to 161 countries and only 8 of these countries are being incorporated by the author, the country Australia is one of these 8 and this is one of the cases where differencing of the second order had to be applied.(Okunogbe et al. 2021)
For a more detailed description, of how the model was fitted exactly, please find an example code for the group 'Boys'.

INSERT TABLE OF A FEW ESTIMATES

The prevalence of overweight and obesity can now be used for various calculations like computing the population with overweight or obesity in absolute numbers. However, before we move on to direct and indirect costs in which calculations like these are required, we use the overweight and obesity prevalence to estimate the "overweight and obesity attributable fraction".

Overweight and obesity attributable fraction

The overweight and obesity attributable fraction (OAF) is the proportion of total health expenditure that can be attributed to overweight and obesity. Following the approach of the reference paper, data on OAO for 52 countries is sourced from the OECD Report (OECD 2019), more specifically from the Global Burden of Disease study. This study analyzes mortality on numerous different diseases, including 28 diseases which are evidently linked to overweight and obesity. The OAF values are presented as an average between 2020-2050 but unfortunatey only represent 52 countries. It is therefore necessary to derive OAF values for countries not included in this study but very well included in this paper.

This is done by regressing the available OAFs of the 52 nations from the OECD study on each country's average overweight and obesity prevalence between 2020 and 2050. The average overweight and obesity is being calculated when predicting future values for overweight and obesity as shown in the Appendix. Thus, this is done group wise just as it is when fitting the autoregressive models to predict future prevalence values. The weighted mean of these average prevalence is then calculated by multiplying each data point value by its weight. Weights are proxied by population percentages.
In the end, one obtains the average prevalence in 2020-2050 for each country. The weighted mean of the 52 countries included in the Global Burden of Disease study is used when the regression of the available OAFs is done on these average prevalences.

By fitting a simple linear regression, we obtain a coefficient which can be used to estimate OAF for countries not included in the report and also fore predicting future OAF values.
Similar to the resulting model by Okunogbe A et al., there is a significant positive association between OAF and obesity prevalence ($\beta = 0.0889, p = 0.0351$)(Figure 1)

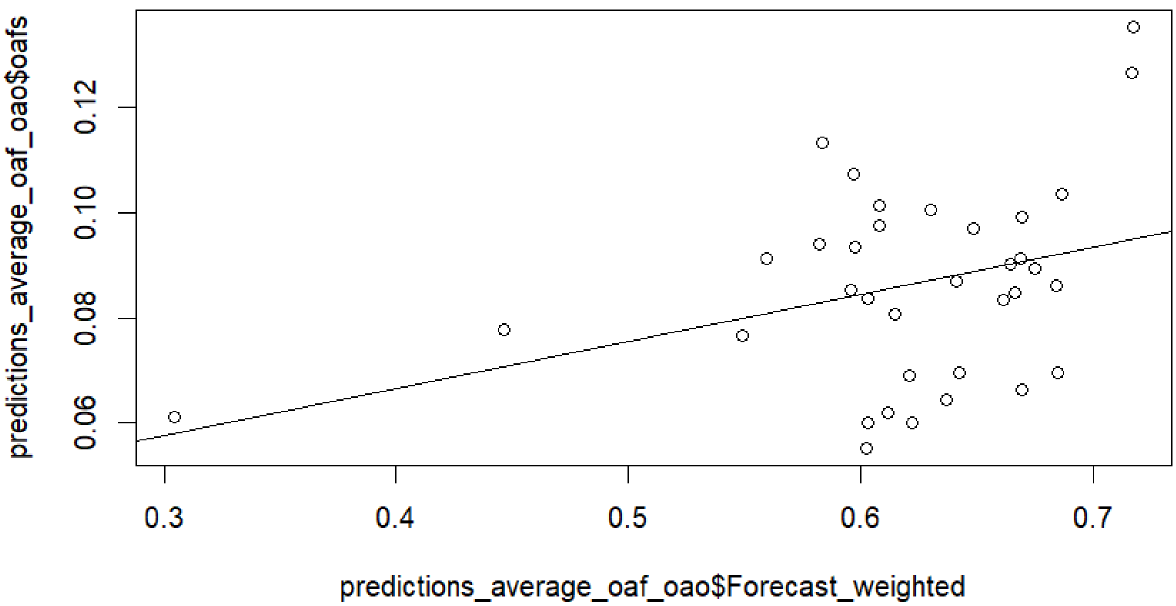


Figure 1: Figure 1

To ensure consistency, the estimated OAFs from the regression output are always used instead of the raw values form the OECD report. For example, when calculating direct medical costs as we will see in the susequent chapter, not the the OAF values from the OECD report are being used, but the OAF values which we predicted

Direct costs

Direct costs are the total medical costs associated with treating illnesses linked to overweight and obesity. This covers medical resources and non-medical resources associated with treating these diseases and also non-medical resources. The diseases considered are the ones which are used in the Global Burden of Disease Study in terms of OAFs.

Direct medical costs

These are expenditures that are directly attributable to patient care and are incurred in the diagnosis and treatment of a specific illness or health condition. These costs cover various aspects of healthcare services such as hospital care, drugs, visits to clinics, and medical interventions. They are essential components of the overall economic impact of a health condition as they play a crucial role in healthcare planning, resource allocation, and economic evaluations

Direct medical costs are calculated by using the OAFs of each country as mentioned earlier. More specifically, data on total health expenditure is available for most countries from the WHO Global Health Expenditure Database. These values can then, country,- and year-wise, be multiplied by the proportion of health expenditures that are attributable to the the overweight and obesity related diseases(OAF). This results into the following formula:

$$DMC = OAF \times TotalHealthExpenditure$$

However, the dataset on total health expenditure unfortunately only includes data until the year 2050 which sets a limit when predicting future direct medical costs. The calculations for direct medical costs are therefore only made for the years 2019-2050. Results of direct medical costs are very similiar to the ones in the reference paper which can be seen in the appendix. Direct medical costs ranged from ... in ... to ... in ... . It can very well be seen how these expenditures play a big role in total expenditures related costs overweight and obesity.

INSERT TABLE OR GRAPH (of absolute values) INSERT PIECHART (compared to other expenses)

Direct non-medical costs

Extra expenses which are not directly tied to patient care but still incur in the process of seeking medical attention are measured by direct non-medical expenditures. This might include the cost of transportation to medical facilities or doctor's visits, the costs paid to carers, the cost of housing and meals while receiving inpatient treatment, and the cost of making home adaptations. An estimate of the estimated travel and carer expenditures for individuals with obesity is included in this study.

Travel costs

We start by calculating travel costs. This is done for inpatient (hospitalization) and outpatient care separately. Inpatient care is the care of patients that involves admission to a hospital. Outpatient care on the other care is anything that does not require hospitalization.

Travel costs are computed by multiplying the average transport cost per trip by the population's obesity rate and the number of additional inpatient or outpatient visits relative to the healthy population.

$$\text{Inpatient Travel Costs} = \text{ATC} \times \text{N}_{\text{in}} \times \text{Population with Obesity}$$

$$\text{Outpatient Travel Costs} = \text{ATC} \times \text{N}_{\text{out}} \times \text{Population with Obesity}$$

where:

- ATC represents the average travel cost to and from health facility
- N_{out} is the number of additional outpatient visits to a health care facility when overweight or obese
- N_{in} is the number of additional inpatient visits to a health care facility when overweight or obese

- Population with Obesity is the overweight or obese population

The average transport cost is proxied by the daily average transport expenditure per capita for each nation. To estimate the average transport cost, the cost of gasoline per liter is and using a global estimate of 5km as an average distance to a health facility and 0.06l/km as average fuel efficiency. Data on gas prices is sourced from the World Development Indicators databank from the World Bank. Transport costs are assumed to stay constant for calculations of future costs.

$$\text{ATC} = \text{Price per Liter of Gasoline} \times \text{Distance to Health Facility} \times \text{Fuel Efficiency}$$

Following the approach in the reference paper, global estimates for inpatient and outpatient visits per country are used. However, as these proxies are sourced from peer reviewed studies and there is no detail on which peer reviewed studies exactly are being used, this paper uses income-level estimates for inpatient and outpatient visits. Countries are divided into the income groups “low income (LIC)”, “lower middle income(LMIC)”, “upper/high middle income(HMIC)” and “high income(HIC)” using Worldbank´s data on which countries belong to which income groups.
The number of additional inpatient and outpatient visits taken as an estimation for HIC is 0.01 and 1.2 sourced from . . .
For other income levels we used . . . as estimates which are 0.16 for outpatient visits and 0.06 for inpatient visits. Average transport costs These estimates are assumed to stay constant when incorporating these constants in calculation for future costs.

Data on absolute numbers of population is sourced from United Nation´s World Population Divisions Database. These include country-level estimates until 2060. We thus applied these numbers to our predictions of overweight and obesity prevalence, yielding the total number of overweight and obese people within each country.

$$\text{Travel Costs} = \text{Inpatient Travel Costs} + \text{Outpatient Travel Costs}$$

Summing the outpatient and inpatient travel Costs up to obtain total direct costs is done on a country level basis and for each year until 2060. Outputs are similiar to the results of Okunogbe et al.

Informal Caregiver (ICG) costs

The last part of direct costs are informal caregiver costs. This includes travel cost of informal caregivers while assuming one informal caregiver per patient. Time costs are assumed to be included within the indirect costs from absenteeism which will be calculated in the subsequent section. Furthermore, to ensure consistency with the reference paper, ICG costs are only taken into account for impatient care. After considering these factors, ICG Costs are calculated by using $\text{ATC} \times \text{N}_{\text{in}} \times \text{Population with Obesity}$ just as when calculating inpatient travel costs.

Indirect costs

We have now covered the most relevant aspects in terms of direct costs. We therefore move on to indirect costs, which refer to the expenses due to premature mortality and productivity losses. Productivity losses can be furthermore divided into Absenteeism which are basically the and Presentteism Cost

Economic costs of premature mortality

Economic costs of premature mortality is what we consider the economic costs of lives lost prematurely due to overweight and obesity attributable deaths. This computation consists of multiplying the total years of potential lives lost (YLL) among individuals who died due to overweight or obesity-related causes, by the

economic value assigned to a life year.
There are multiple possible proxies for economic value of a life year such as annual wages or value of statistical life. But we ultimately decided to stick with GDP per capita as it recognizes all economic contributions regardless of job status. This adds an equitable perspective to the computation. This analysis furthermore extends this approach by incorporating a GDP multiplier to capture the effects associated with changes in national income, mortality or life expectancy. In alignment with the reference paper we apply a multiplier of *1.6* to each estimate as a global estimate.

1) Data on deaths caused due to overweight and obesity related diseases are sourced from the Global Burden of Disease Study 2019. As this study includes total deaths of various different diseases we filter out the 28 relevant diseases.
The goal is to calculate the amount of deaths which are solely attributable to overweight and obesity. This is done by incorporating the relative risk and then calculating the population attributable fraction from this. In this context, relative risk is a measure quantifying the likelihood of a particular disease occurring in a population without overweight or obesity compared to the overweight and obese population. The relative risk values are obtained from the GBD Study 2019.
From this, the Population attributable fraction (PAF) can be calculated to measure the proportion of deaths that are due to overweight and obesity. It is calculated as:

$$PAF = \frac{(p \times (RR - 1)) + 1}{p \times (RR - 1)}$$

Where:

- p is the prevalence of overweight and obesity in the population.
- RR is the relative risk.

By multiplying the PAFs with all-cause mortality of the relevant diseases, we ultimately get the deaths of obesity related diseases which are evidently linked to overweight and obesity. This number is ultimately multiplied by the value of a life year. The described calculations are done age-group (a), sex(s) and cause/disease(c) wise and can be summarized by the following formula:

$$\sum_{s,a,c,t} \text{Overweight and Obesity related deaths} = \sum_{s,a,c,t} PAF \times \sum_{s,a,c,t} \text{Total deaths}$$

After obtaining the amount of deaths which are attributable to overweight and obesity the costs of these deaths can be calculated by multiplying this number by the value of a life year. In other words, the This way, we basically compute

2) It is important to note that the approach of calculating premature mortality costs as described below might not align fully with the reference paper´s approach. This done because the alternations make calculations smoother and easier to follow and due to missing data.

However, the main aspect of the paper´s approach of using data of overweight and obesity related diseases from the Global Burden of Disease Study 2019 is also applied in this thesis. As the GDB study includes total deaths and Years of lives lost(YLLs) of various different diseases we filter out the 28 relevant diseases to start calculations. Following the approach of Okunugbe et al, the first step in determining premature mortality costs is getting the YLLs due to overweight and obesity related diseases. Just like in the reference paper, YLLs are at the core of calculating premature mortality costs. This aspect is therefore covered in detail below:

To make calculations easier, we directly used the YLLs values of the Global Burden of Disease Study 2019. These are not being used by Okunugbe et al. In the reference paper, YLLs are calculated by using life

expectancy sourced from the World Population Prospect database.
This makes the process of determining premature mortality costs much smoother and avoids unnecessary calculations. However, it is important to mention that reconstructing premature mortality costs by directly using the available YLLs values from the GDB Study 2019 makes the results slightly different to the results we obtained when using the original approach.

After obtaining the YLLs of overweight and obesity related diseases, the goal is to calculate the YLLs which are solely attributable to overweight and obesity. This can be illustrated with an example: There are 1 Million deaths of the overweight and obesity related disease *Diabetes*. However a lot of theses deaths are attributable to different factors other than overweight and obesity.

YLLs attributable to overweight and obesity are calculated by incorporating the relative risk and then deriving the population attributable fraction from this.
In this context, relative risk is a measure quantifying the likelihood of a particular disease occurring in a population without overweight or obesity compared to the overweight and obese population. The relative risk values are obtained from the GBD Study 2019. Sadly, we were only available to get the relative risks for the base year 2019, which brings us to the second marking difference in the approach of calculating premature mortality costs:
The reference paper uses relative risks projections for 2019-2060, making it possible to calculate the resulting YLLs for future years. As this is not possible for us, this paper primarily calculates the YLLs and costs for the base year 2019 and predictions are made later and described subsequently.

As already mentioned, by incorporating relative risks, the population attributable fraction (PAF) can be calculated to measure the proportion of YLLs that are due to overweight and obesity. It is calculated as:

$$PAF = \frac{(p \times (RR - 1)) + 1}{p \times (RR - 1)}$$

Where:

- p is the prevalence of overweight and obesity in the population.
- RR is the relative risk.

By multiplying the PAFs with the number of years of lives lost of the relevant diseases, we ultimately get the years of lives lost which are evidently linked to overweight and obesity. This number is ultimately multiplied by the value of a life year. The described calculations are done age-group (a), sex(s) and cause/disease(c) wise and can be summarized by the following formula:

$$\sum_{s,a,c} \text{Ylls due to overweight and obesity} = \sum_{s,a,c} PAF \times \sum_{s,a,c} \text{Total Ylls}$$

After obtaining the Years of lives lost which are attributable to overweight and obesity, the costs of these deaths can be calculated by multiplying it by the value of a life year. To make things more understandable, we take Australia as an example and see the different values of the crucial parameters.
As described, mortality cost is obtained by multipliying the Ylls by the value of a life year (proxied by GDP/capita) and the GDP multiplier.
It is important to note that these are only the costs for the year 2019. We will now apply a simple method of projecting these into the future.

INSERT TABLE + CODE OF AUSTRALIA

Projections of ##reduced productivity due to additional absenteeism:

##reduced productivity due to additional presenteeism:

Contribution

Question

Structure

Findings/Discussion

Conclusion

Appendix

forecasting prevalence of men

```
library(forecast)

forecast_df_men <- data.frame()
m_2020_2050_men <- data.frame()
for (country in unique(urban_per$Location)) {
  country_df <- data_oao_men[data_oao_men$Country.Region.World == country, ]
  ag_popu_df <- ag_population_adult[ag_population_adult$Location == country, ]
  ag_popu_df2<-ag_popu_df[ag_popu_df$Year>=2019, ]
  ag_popu_df<-ag_popu_df[ag_popu_df$Year<=2016, ]
  print(country)

  # Extract the time series and exogenous variables
  country_df$oao_logit <- log(country_df$oao / (1-country_df$oao))
  country_df$oao_log <- log(country_df$oao)

  ts_data <- ts(country_df$oao_logit, start = min(country_df$Year), frequency = 1)
  xreg_data <-ag_popu_df$PopMale
  xreg_data2<-ag_popu_df2$PopMale


  # Initialize arimax_model outside the tryCatch block
  arimax_model <- NULL

  tryCatch({
    # Try fitting ARIMAX model
    arimax_model <- Arima(ts_data, order = c(2, 1, 0), xreg = xreg_data)
  }, error = function(e) {
    # If an error occurs, print the error message
    print(paste(country, "Error1:", e))
  })

  if (!is.null(arimax_model)) {
    # Generate forecasts using the fitted model
    forecasts <- forecast(arimax_model, xreg = xreg_data2, h = 42)
```

```
    forecasts_original_scale <- exp(forecasts$mean) / (1 + exp(forecasts$mean))
    #forecasts_original_scale <- exp(forecasts$mean)

    # Create a data frame for forecasted values
    forecast_df_country <- data.frame(Country = rep(country, 42), Year = seq(2019, 2060), Forecast_men =
    forecast_df_men <- rbind(forecast_df_men, forecast_df_country)

    # Calculate the mean projection for 2020-2060
    country_mean <- mean(forecasts_original_scale[1:32])  # Adjust to use values for the years 2020-205
    m_2020_2050_men <- rbind(m_2020_2050_men, data.frame(Country = country, MeanProjection = country_mea
  }
 if (is.null(arimax_model)) {
  tryCatch({
    arimax_model <- Arima(ts_data, order = c(2, 2, 0), xreg = xreg_data)
    # Generate forecasts using the fitted model
    forecasts <- forecast(arimax_model, xreg = xreg_data2, h = 42)
    forecasts_original_scale <- exp(forecasts$mean) / (1 + exp(forecasts$mean))
    #forecasts_original_scale <- exp(forecasts$mean)

    # Create a data frame for forecasted values
    forecast_df_country <- data.frame(Country = rep(country, 42), Year = seq(2019, 2060), Forecast_men =
    forecast_df_men <- rbind(forecast_df_men, forecast_df_country)

    # Calculate the mean projection for 2020-2060
    country_mean <- mean(forecasts_original_scale[1:32])  # Adjust to use values for the years 2020-205
    m_2020_2050_men <- rbind(m_2020_2050_men, data.frame(Country = country, MeanProjection = country_mea
  }, error = function(e) {
    # Print an error message if the code block fails
    print(paste("Error2 in block:", e))
  })
  }
}


# Print the results
head(forecast_df_men)
print(m_2020_2050_men)
```

“‘

References

OECD. 2019. *The Heavy Burden of Obesity*. https://doi.org/https://doi.org/https://doi.org/10.1787/67450d67-en.

Okunogbe, Adeyemi, Rachel Nugent, Garrison Spencer, Johanna Ralston, and John Wilding. 2021. “Economic Impacts of Overweight and Obesity: Current and Future Estimates for Eight Countries.” *BMJ Global Health* 6 (10). https://doi.org/10.1136/bmjgh-2021-006351.